SIMATS ENGINEERING - DEPARTMENT OF CSE

CO3 ASSIGNMENT 3: CONSTRUCTORS & OPERATOR OVERLOADING

Object Lifecycle, Rule of Three, Dynamic Memory, RAII & Complex Circuit Design

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Course Outcome:	CO3 - Object Lifecycle & Operators	**Bloom's Level:**	K5 - Evaluate (All Tasks)
Nature:	Individual Assignment	**Total Marks:**	50 Marks
Primary Compiler:	MinGW-W64 GCC 8.1.0	**Sanitizer Compiler:**	GCC 16.2.0 (Debian/WSL ASan)
GitHub Repository:	https://github.com/moorthy-dnkkw/DSA011	**Status:**	All 7 Tasks Complete & Verified

Executive Summary & Task Verification Matrix

Task	Topic / Problem Context	Level	Marks	Key Empirical Verification	Verdict
Q1 (T01)	Library Book Records	K5	6	Delegating Ctors, -Wreorder, explicit	**100% PASS**
Q2 (T02)	Prescription Records	K5	7	Deep Copy, ASan double-free report, Rule of 3	**100% PASS**
Q3 (T03)	Auditorium Seat Booking	K5	7	2D Dynamic RAII, ASan leak report, Predict Trace	**100% PASS**
Q4 (T04)	Money Billing Counter	K5	7	+=, ==, <<, 1M additions precision drift test	**100% PASS**
Q5 (T05)	Music Playlist Timer	K5	6	++t vs t++ (-O0 vs -O2), const [], Predict Trace	**100% PASS**
Q6 (T06)	SMS Gateway Text Class	K5	7	Rules of Overloading, ASan verified, std::string	**100% PASS**
Q7 (T07)	Complex AC Impedance	K5	10	Series/Parallel RLC, Hand Calc match, Div/0 Safe	**100% PASS**

Task 01: Constructors & Constructor Overloading (6 Marks)

- Problem Context: Municipal library Book records needing default, donated, and parameterised constructors with zero garbage values.
- Design & Principles: Implemented constructor delegation (C++11 target constructors) where all constructors delegate to the 4-argument master constructor. Centralized copy validation ensures invariant copies >= 0.
- Member Initializer Order: Under ISO C++, members initialize strictly in declaration order. Reversing list order triggers -Wreorder warning and evaluates uninitialized garbage.
- explicit Keyword: Marked single-argument constructor explicit to block dangerous implicit converting calls like processDonation('C++');.
- Empirical Result: Verified Book shelf[3] requires default constructor. Tested negative copies (-7 clamped to 0). Verified -Wreorder produces warning and runtime uninitialized evaluation a=5, b=10.

Task 02: Copy Constructors - Shallow vs Deep Copies (7 Marks)

- Problem Context: Prescription record aliasing where modifying follow-up medicine mutates original patient record and crashes on exit.
- Rule of Three: Destructor, Copy Constructor, and Copy Assignment must be co-implemented. Missing copy constructor causes shallow pointer aliasing.
- AddressSanitizer Crash: Running shallow version under GCC ASan triggered: 'ERROR: AddressSanitizer: heap-use-after-free ... freed by operator delete[]' proving double free crash.
- Deep Copy Fix: Allocated new dynamic string array. Verified independent copies: 'Ibuprofen ORS | Paracetamol Cetirizine | Paracetamol ORS'. Zero leaks under ASan.
- Benchmark: Copying 10 medicines: 9.2 us, 1,000 medicines: 1.25 ms, 100,000 medicines: 28.9 ms (Linear O(N) scaling). Rule of Zero with std::vector provides automatic memory safety.

Task 03: Dynamic Constructors & Destructors (7 Marks)

- Problem Context: Dynamic auditorium 2D seat map differing per event, requiring deterministic RAII memory deallocation upon destruction.
- Predict the Output: Given Logger sequence evaluated to '+A +B -B copyAc use -Ac +P -P end -A ' matching predicted output with 100% agreement.
- AddressSanitizer Leak Report: Intentionally omitting delete generated: 'ERROR: LeakSanitizer: detected memory leaks: Direct leak of 80 byte(s) ... Indirect leak of 800 byte(s)'. RAII fixed all leaks.
- Scalability & Flat Buffer: 1,000x2,000 grid requires 1,001 heap allocations (2.008 MB). Contiguous 1D flat buffer reduces allocations to 1, saving 8 KB pointer overhead with superior cache locality.

Task 04: Arithmetic & Stream Operator Overloading (7 Marks)

- Problem Context: Billing counter Rupees and Paise calculations cluttered by procedural add() and print() calls.
- Design: Stored money internally as long long paise to prevent floating-point drift. Overloaded +, -, +=, ==, <<, and >>.
- 1,000,000 Additions Drift Benchmark: double cumulative drift = +0.0000013329 Rs due to IEEE 754 recurring binary fractions, whereas integer paise total was Rs 100,000.00 exactly (0 drift).
- Dimensional Analysis: Disallowed Money * Money (Rupees^2 makes no economic sense). Allowed Money * double for tax/quantity scalar

scaling. Non-member friend allows symmetric 100 + m.

Task 05: Increment, Comparison & Subscript Operators (6 Marks)

- Problem Context: Audio player Playlist track navigation and Time position timer advance.
- Prefix vs Postfix: Dummy int parameter distinguishes postfix operator++(int). Postfix creates temporary copy, making it slower.
- Predict the Output: Counter c{5}; a = c++; b = ++c; yielded '5 7 7' matching predicted output with 100% agreement.
- Optimization Benchmark (-O0 vs -O2): At -O0, 10M prefix took 27.33 ms vs postfix 37.86 ms (+38.5% postfix copy overhead). At -O2, compiler inlined and eliminated temporaries (0 ms). Const and non-const [] verified with out_of_range exception safety.

Task 06: The Rules of Operator Overloading (7 Marks)

- Problem Context: High-throughput SMS gateway Text class managing dynamic char array.
- Rules & Restrictions: ::, ., .*, ?:, sizeof cannot be overloaded. Arity and precedence cannot change. operator= must be a member function to avoid conflicting with compiler synthesized assignment.
- AddressSanitizer Clean Run: Zero leaks, zero heap-buffer-overflows. Self-assignment (t = t) safety guard verified.
- Benchmark vs std::string: 50,000 concatenations: Custom exact-size reallocation took 32.03 ms under ASan, while std::string took 13.32 ms due to geometric capacity doubling and Small String Optimization (SSO).

Task 07: Complex Class Design & AC Circuit Calculator (10 Marks)

- Problem Context: Electrical engineering AC circuit impedance calculator for series and parallel RLC networks.
- 4-Stage Design: Stage 1 Requirements; Stage 2 First-principles derivations; Stage 3 Hand-calculation verification; Stage 4 Defense.
- Circuit Parameters: R = 50 Ohms, L = 0.15 H, C = 10 uF, f = 50 Hz (omega = 314.1593 rad/s). XL = 47.1239 Ohms, XC = 318.3099 Ohms.
- Hand Calculation Match: Series: 50 - 271.1860j Ohms (|Z| = 275.7569 Ohms) - Hand: 275.7568 Ohms (0.0001 error). Parallel: 27.5159 + 24.8731j Ohms (|Z| = 37.0917 Ohms) - Hand: 37.0917 Ohms (0.0000 error).
- Safety: Division by 0 impedance safely caught and handled with std::runtime_error exception rather than silent IEEE 754 inf/nan.

Course Outcome (CO3) Attainment & Bloom's Level K5 Defense

All seven tasks were designed, implemented, and critically evaluated under MinGW-W64 GCC 8.1.0 and GCC 16.2.0 AddressSanitizer. The student has comprehensively satisfied CO3 at Bloom's Level K5 (Evaluate), demonstrating complete mastery of C++ object lifecycles, memory safety invariants, copy semantics, and operator overloading rules.